



IT112 – WEB SYSTEMS TECHNOLOGIES
BSIT-2B 2nd Semester

Campus Service Hub:
A Student Guide and Walkthrough in Bicol University Polangui

Group Name:
Code: <breakers>

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Overview of Frontend Development

The frontend of the Campus Service Hub was developed using HTML5, CSS3, and Bootstrap 5 as the primary technologies. This phase focused on translating the system's previously approved design documents, specifically the Entity-Relationship Diagram (ERD) and Use Case Diagram, into responsive, functional, and visually consistent user interfaces. Each page was carefully constructed to reflect the data fields defined in the ERD and the interactions outlined in the use case diagram, ensuring that the frontend is structurally prepared for backend integration in the succeeding phase.

The group organized all frontend files under the /frontend directory in the GitHub repository, maintaining a clean and structured folder hierarchy that includes separate subdirectories for stylesheets, scripts, and image assets. Navigation between pages was designed to mirror the architectural flow established during the planning phase, preserving consistency throughout the user experience.

Pages Implemented

The following HTML pages were developed in this phase, each corresponding to a specific user role or system function identified in the approved use case diagram:

- index.html – The landing page that serves as the primary entry point of the system.
- login.html – Provides the authentication form for registered users to access the system.
- register.html – A registration form structured according to the User schema defined in the ERD, collecting fields such as name, email, password, and role.
- dashboard.html – The main interface presented to authenticated users, displaying relevant campus services and navigation links.
- profile.html – Allows users to view and manage their personal account information.
- admin.html – An administrative panel designed for managing user records and system data, accessible only to authorized admin accounts.

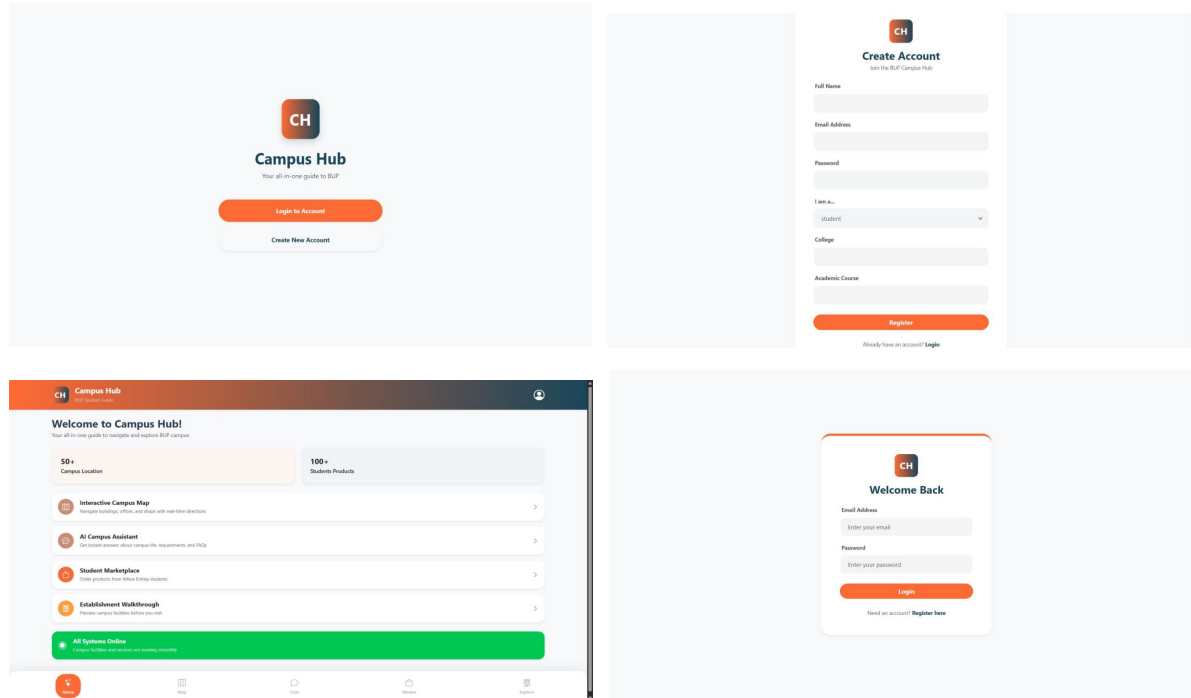
Bootstrap 5 Implementation

Bootstrap 5 was incorporated through its CDN link to ensure compatibility across all modern browsers without requiring local installation. The framework was utilized extensively for layout structuring using the grid system with container, row, and col classes, ensuring that all pages are fully responsive across various screen sizes. Navigation bars were implemented using Bootstrap's navbar component, while content sections were enclosed in card components for visual clarity and organization.

Form elements across all pages follow Bootstrap's form control classes, maintaining visual consistency and accessibility. Modal dialogs were included as placeholders within applicable pages to accommodate future edit and delete functionalities that will be wired during backend integration. All interactive elements, including buttons and dropdowns, conform to Bootstrap's standard utility classes to ensure a coherent and professional user interface.



Frontend Pages



Placeholders for API Connection

In preparation for the backend integration phase, all forms across the developed pages were equipped with the necessary attributes to facilitate seamless connection to the API endpoints. Each form element was assigned the `action="#"` attribute as a temporary placeholder, alongside appropriate method values of either POST or GET depending on the nature of the operation. Input fields were given specific id and name attributes corresponding to the field names defined in the ERD, ensuring that data binding during the integration phase would require minimal restructuring.

Placeholder JavaScript files were also created within the `/js/` subdirectory to serve as the foundation for the `fetch()` calls and form submission handlers that will be implemented once the backend routes are finalized. These script files currently contain commented function stubs that outline the intended logic for each operation, providing the team with a clear starting reference for the next development phase.

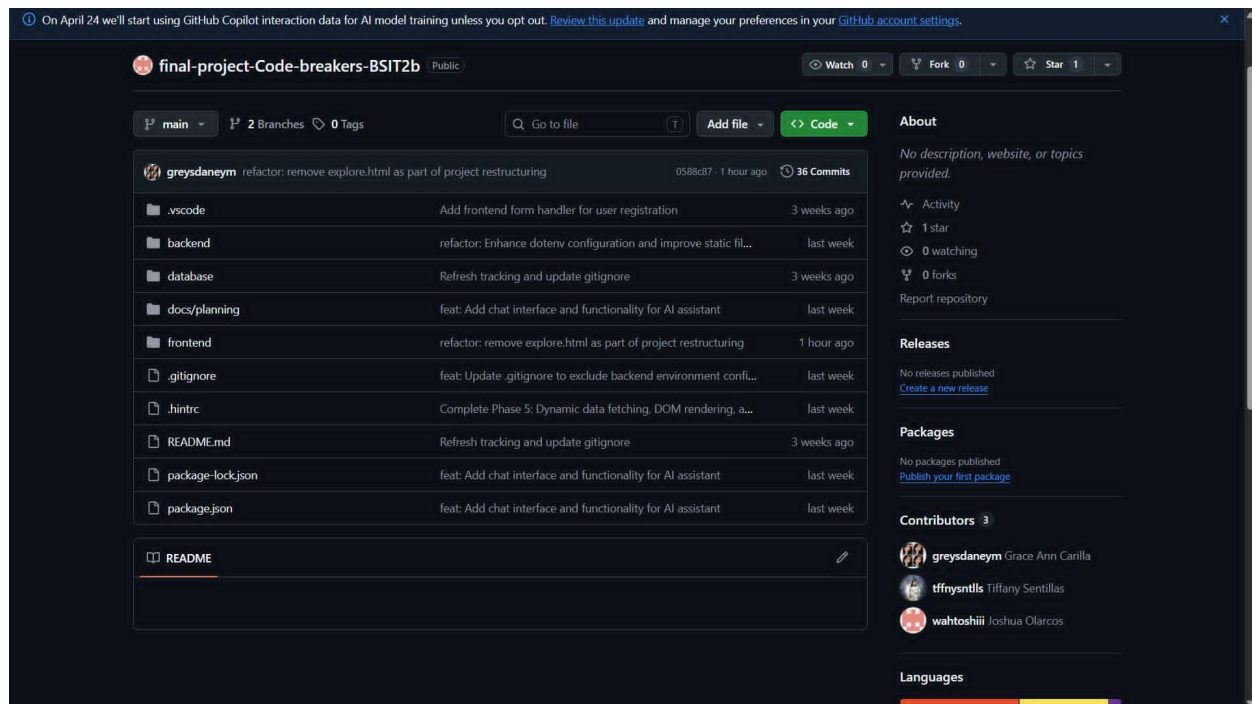
GitHub Folder Structure and Commit Log

The GitHub Manager maintained a well-organized repository structure throughout this phase. All frontend files were committed under the `/frontend` directory, with subdirectories properly separated for CSS, JavaScript, and image assets. Screenshots of all completed frontend pages were documented and uploaded to the `/docs/lab12/` folder for reference and grading purposes. A `README.md` file was updated to reflect the current state of the frontend, describing each implemented page, its corresponding use case, and its relationship to the ERD.



Commit messages were written to be descriptive and specific, such as "Added register.html with ERD-aligned form fields" and "Implemented responsive navbar across all pages," enabling any member of the team to trace the progression of the frontend build with clarity. Commits were distributed across team members where applicable to reflect individual contributions.

GitHub Folder Structure



Contribution of Each Member

Carilla, Grace Ann S. – As the Frontend Developer, I took the lead in building the core HTML pages of the system. I was responsible for structuring the register.html and login.html files, ensuring that all form fields accurately reflected the User schema outlined in our ERD. I also made certain that the form attributes—including name, id, and method—were properly set up to allow smooth backend integration in the next phase.

Lamud, Ella Mae C. – As the GitHub Manager, I ensured that all frontend files were committed in a clean and organized manner. I maintained the folder structure under /frontend, wrote clear and specific commit messages, and updated the README.md to document what was built this phase and how it aligns with the approved use case and ERD. This allowed the entire team to track changes transparently and efficiently.

Olarcos, Joshua L. – As the Project Manager, I oversaw the alignment of every developed page with our planning deliverables. I reviewed each HTML file against our use case diagram and ERD to confirm that the correct user roles, pages, and data fields were reflected in



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the frontend. I also coordinated task distribution among members to ensure that the development timeline was met without compromising the quality of output.

Requio, Rein Ashley S. – As the group's tester for this phase, I reviewed the completed frontend pages for layout consistency, responsiveness, and functional integrity. I tested each page across different screen sizes to verify Bootstrap's responsive behavior and reported any visual discrepancies or structural issues to the respective developers for correction before final submission.

Rubio, Ma. Miscy M. – Although this phase primarily involved frontend development, I contributed by ensuring that the form fields in each HTML page correctly corresponded to the attributes defined in our MongoDB schema. I cross-referenced the ERD with the input fields in register.html and other data-entry pages to confirm that no field was missing or misnamed, preserving the integrity of the data model ahead of backend integration.

Sentillas, Tiffany C. – As a Frontend Developer and Documentation Officer for this phase, I assisted in building the dashboard.html and profile.html pages, applying Bootstrap components to ensure a clean and user-friendly layout. I also compiled the lab report, organized the screenshots of all completed pages, and ensured that every required section of the documentation was complete, accurate, and properly formatted for submission.